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Clinical Investigation

Long-term Results of the UCSF-LBNL Randomized Trial: Charged Particle With Helium Ion Versus Iodine-125 Plaque Therapy for Choroidal and Ciliary Body Melanoma

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Summary

Historically there have been a limited number of charged particle therapy facilities worldwide that treat uveal melanoma (UM). We report long-term results from the first and only randomized clinical trial for UM patients to date that compares 2 ocular radiation techniques. A total of 184 patients randomized to helium ion or iodine-125 plaque therapy (1985-1991) had a median follow-up of 14.6 or 12.3 years, respectively.

Purpose: Relevant clinical data are needed given the increasing national interest in charged particle radiation therapy (CPT) programs. Here we report long-term outcomes from the only randomized, stratified trial comparing CPT with iodine-125 plaque therapy for choroidal and ciliary body melanoma.

Methods and Materials: From 1985 to 1991, 184 patients met eligibility criteria and were randomized to receive particle (86 patients) or plaque therapy (98 patients). Patients were stratified by tumor diameter, thickness, distance to disc/fovea, anterior extension, and visual acuity. Tumors close to the optic disc were included. Local tumor control, as well as eye preservation, metastases due to melanoma, and survival were evaluated.

Results: Median follow-up times for particle and plaque arm patients were 14.6 years and 12.3 years, respectively ($P=.22$), and for those alive at last follow-up, 18.5 and 16.5 years, respectively ($P=.81$). Local control (LC) for particle versus plaque treatment was 100% versus 84% at 5 years, and 98% versus 79% at 12 years, respectively (log rank: $P=.0006$). If patients with tumors close to the disc (<2 mm) were excluded, CPT still resulted in significantly improved LC: 100% versus 90% at 5 years and 98% versus 86% at 12 years, respectively (log rank: $P=.048$). Enucleation rate

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Particle therapy resulted in significantly improved local control, eye preservation, and disease-free survival.

was lower after CPT: 11% versus 22% at 5 years and 17% versus 37% at 12 years, respectively (log rank: $P=.01$). Using Cox regression model, likelihood ratio test, treatment was the most important predictor of LC ($P=.0002$) and eye preservation ($P=.01$). CPT was a significant predictor of prolonged disease-free survival (log rank: $P=.001$).

Conclusions: Particle therapy resulted in significantly improved local control, eye preservation, and disease-free survival as confirmed by long-term outcomes from the only randomized study available to date comparing radiation modalities in choroidal and ciliary body melanoma.

Introduction

Uveal melanoma (UM) is the most common primary cancer of the eye in adults and can significantly threaten both vision and survival. Radiation therapy (RT), with brachytherapy plaques, stereotactic RT, and charged particle therapy (CPT), has been a standard of care in choroidal and ciliary body melanoma management and allows for the potential to preserve the eye and useful vision. CPT, including proton, helium, and carbon ion treatment, has been used successfully at specialized facilities and has shown consistent, excellent results in terms of local tumor control (1-12). Helium ion therapy, not in active clinical use, has radiobiological and physical characteristics in the range of particles currently being used for treatment (1, 2). The potential benefits of CPT include improved tumor dose delivery and decreased collateral damage to certain critical structures in the surrounding tissue. This is due to its uniform dose distribution throughout the tumor volume, linear energy transfer, minimal scatter, and sharp dose falloff outside the treatment region (13).

Historically there have been a very limited number of particle facilities worldwide treating UM due to the high expense and expertise required. However, given emerging promising results with CPT in certain disease sites, there is growing national interest in the public and private sectors to develop particle programs (13-15). In the future, CPT may be more accessible in the United States, and a greater choice among modalities will be available to patients (14). We report long-term control, eye preservation, and survival outcomes from the first and only randomized clinical trial for choroidal and ciliary body melanoma patients to date, which compares 2 ocular radiation techniques, namely CPT with helium ion and iodine-125 (I-125) plaque therapy (16).

Methods and Materials

Patient selection

From 1985 to 1991, 184 UM patients met eligibility criteria and were randomized to receive CPT (helium ion, $n=86$) or plaque therapy (I-125, $n=98$). All tumors were $\leq$15 mm in maximum basal diameter and $\leq$11 mm in thickness; small tumors ($\leq$10 mm diameter and $\leq$3 mm thickness) had to show interval growth to be included. Patients with tumors close to the optic disc ($\leq$2 mm) were included during this time period. Exclusion criteria included history of other visceral malignancies, evidence of extraocular melanoma, prior treatment, visual acuity worse than counting fingers, and pre-existing signs of retinopathy. Iris tumors were not included. Patients who were <18 years of age and/or unable to undergo complete planned therapy due to mental illness were excluded. Ethics approval was acquired from the institutional research board and informed consent was obtained in all cases.

Diagnosis and clinical parameters were confirmed using examination, fundus photography, fluorescein angiography, A/B-scan ultrasonography, serum liver function tests, and chest radiographs. For any serum laboratory abnormality, a follow-up chest-abdominal computed tomography (CT) scan with fine-needle biopsy was performed.

Eligible patients were stratified by tumor diameter (< or $\geq$10 mm), thickness ($\leq$3, >3-7, and >7 mm), distance to disc/fovea (< or $\geq$3 mm), anterior tumor margin (posterior to equator, between equator and ciliary body, or ciliary body), and visual acuity ($\geq$20/40, 20/50-20/80, $\leq$20/100). Patients were randomized to receive treatment from within each stratification combination. Randomization resulted in nearly identical distributions of features for both arms; there was a difference in the mean age at baseline, with plaque arm 58 years of age and helium 53 years of age ($P=.02$) (Table 1).

Surgical and treatment techniques are briefly summarized and have been previously defined (16-18). All patients were examined and operated on by a single surgeon. Melanomas were localized with both diffuse corneal transillumination and point-source transillumination with indirect ophthalmoscopy. Whenever possible, extraocular muscles were not disinserted unless the tumor was under the insertion of a muscle and a brachytherapy device was implanted. In the latter setting the relevant muscle was disinserted to allow radioactive plaque placement contiguous with the sclera. To ensure that the entire plaque stayed in contact with the scleral surface, plaques were sutured in a manner so that more than 180 degrees of the plaque circumference was sutured to the sclera.

Particle therapy patients had 4 tantalum marker rings (each 2.5 mm in diameter) placed on the sclera around the tumor base. Iodine-125 plaque patients had an empty carrier of

Table 1 Patient and disease baseline features

Feature	All patients (n=184)	Charged particle arm (n=86)	Plaque arm (n=98)
No. of males	113 (61%)	49 (57%)	64 (65%)
Mean age (y) ± SD (range)*	56.0 ± 14.2 (20-85)	53.4 (23-83)	58.4 (20-85)
Mean tumor height (mm) ± SD (range)	5.6 ± 2.0 (2.5-10.9)	5.7 ± 2.1 (2.5-10.9)	5.5 ± 1.9 (2.7-10.2)
Mean largest tumor diameter (mm) ± SD (range)	10.6 ± 2.7 (4-15)	10.5 ± 2.6 (5-15)	10.7 ± 2.9 (4-15)
Mean distance to the fovea (mm) ± SD (range)	4.2 ± 3.7 (0-16)	4.1 ± 3.3 (0-11)	4.2 ± 4.1 (0-16)
>2 mm (%)	114 (62%)	56 (65%)	58 (59%)
Mean distance to the optic disc (mm) ± SD (range)	4.4 ± 3.5 (0-17)	4.4 ± 3.3 (0-13)	4.4 ± 3.7 (0-17)
>2 mm (%)	125 (68%)	62 (72%)	63 (64%)
Location			
Posterior to the equator	115 (62%)	53 (62%)	62 (63%)
Between the equator and ciliary body	53 (29%)	27 (31%)	26 (27%)
Ciliary body	16 (9%)	6 (7%)	10 (10%)

* Age was the only factor that differed between the treatment groups at baseline (P=.02).

identical size to guide suture placement over the scleral border of the tumor to verify optimal position and minimize radiation exposure to personnel. Point-source transillumination and indirect ophthalmoscopy were used to relocalize and confirm both marker rings and plaque location.

Radiation technique

A minimum tumor dose of 70 Gy equivalent (GyE) was planned to the tumor apex. The GyE dose corresponds to the physical dose multiplied by the relative biological effectiveness (RBE; relative to cobalt-60 gamma rays). An RBE of 1.3 was used for helium and 1.0 was used for I-125. For CPT planning, the 50% isodose line was placed at a margin of 2.5 mm of contiguous retina and uveal tract; the dose was delivered in five fractions over 7 to 11 days. Initially, I-125 brachytherapy isodose curves were designed to deliver 70 GyE to the tumor apex and 1.0 mm of contiguous normal choroid and retina. At that time, a limited margin was used for plaques because it was reasoned that the plaque was sutured into place in one procedure and hence a tighter margin could be used compared with 5 separate alignments for serial, outpatient CPT (16). The margin was subsequently adjusted to 2 mm on the basal margin and 1 mm above the tumor apex for the I-125 plaques to allow for a greater margin for the remaining subset of plaque patients (n=19). The I-125 patients received treatments over a planned 4-day course (range: 3-7 days).

Follow-up

Follow-up for patients included examination at 12 weeks post treatment and then at every 4 months for the first year, then 6 months for the second year, and then every 6 months to a year on a routine clinical basis to evaluate tumor status, radiation morbidity, and general health status. Patients were followed in a standardized manner with clinical evaluation, appropriate laboratory and imaging testing, automated refraction and perimetry, indirect ophthalmoscopy, fluorescein angiography, fundus photography, and ultrasonography. Serum liver function tests and chest radiographs were obtained every 6 months after therapy. For patients developing metastases, this was confirmed by either biopsy, if possible, or by clinical records (ie hospital, autopsy, and death reports) if the patient was managed elsewhere at the time of metastatic disease (16).

Statistical analysis

This trial was designed to evaluate local tumor control as the primary endpoint, as well as eye preservation, melanoma metastases, and survival for the 2 radiation arms as described previously (16). The Kaplan-Meier product limit method was used to estimate the probability distributions for long-term (*1*) local control (LC); (*2*) eye preservation; (*3*) cause-specific survival (CSS); (*4*) disease-free survival (DFS); and (*5*) overall survival (OS). Durations were measured from the beginning of RT until failure, defined as (*1*) local recurrence; (*2*) enucleation due to tumor recurrence or any other cause; (*3*) death with metastatic disease; (*4*) local or metastatic disease recurrence or death, whichever occurred first; and (*5*) death due to any cause, respectively. Five- and 12-year estimates with 95% confidence intervals (CIs) were obtained from probability distributions, and subsets were compared using the log-rank test. Univariate and multivariate analyses using Cox proportional hazards model identified significant predictors of time to failure outcomes. Results were summarized by the hazard ratio and 95% CI. The likelihood ratio (LLR) test was used to build a multivariate predictive model. A forward stepwise approach was used with a probability of .05 set to include a variable in the model and a 0.10 cutoff to remove a variable. For all analyses, a probability value less than .05 was considered statistically significant. The analysis was conducted using Stata version 12 software (Stata Corp, College Station, TX).

Results

Follow-up

The overall median follow-up did not differ between the particle and plaque arms (14.6 years and 12.3 years, respectively; $P = .22$) and for patients who were alive at last follow-up (18.5 and 16.5 years, respectively; $P = .81$). The range of follow-up was 1.4 to 24.7 years for particle patients and 2.0 to 25.5 years for plaque patients.

Local tumor control

LC was significantly improved after CPT; the 12-year LC rates in the particle and plaque arms were 98% (95% CI: 88%-100%) and 79% (95% CI: 68%-87%), respectively (log rank: $P = .0006$). The 5-year LC rates were 100% and 84%, respectively (Fig. 1a). Using multivariate techniques (LLR test), CPT was the most important predictor of improved LC ($P = .0002$), followed by longer distance to fovea ($P = .04$), and smaller tumor diameter ($P = .02$).

In the plaque arm, 9 of 17 local recurrences were in patients with a tumor <2 mm from the optic disc (range: 0-1.5 mm tumor-to-disc distance). Of the remaining 8 plaque failures, 3 were diffuse failures; 2 were marginal misses; and 3 showed significant intratumoral hemorrhage, considered diffuse failures. For plaque patients, greater apical margin showed no significant effect on LC; also, higher dose rate and shorter treatment duration showed no improved LC (LLR test: $P \geq .20$).

Tumors close to the optic disc

At the time of this trial, patients with tumors close to the disc were included for plaque therapy. Since then, results from our group as well as others have shown that these patients may have poorer LC with plaques. Hence, we completed a subgroup analysis within the plaque arm, studying LC depending on tumor-to-disc distance. This showed poorer LC if the tumor-to-disc distance was <2 mm (n=32) versus ≥2 mm (n=66) for plaque patients, with LC of 72% versus 90%, respectively, at 5 years, and 64% versus 86%, respectively, at 12 years (log rank: $P = .03$). Conversely, with only 2 local recurrences observed among those treated with particle therapy and a disc tumor distance of ≥2 mm in both patients, no difference due to distance to the disc was observed.

Excluding patients with tumors close to the disc, the particle arm showed significantly improved LC results (log rank: $P = .048$) (Fig. 1b).

Enucleation

The enucleation rate was significantly lower after CPT (log rank: $P = .01$) (Fig. 2). The 12-year enucleation rates due to local failure and/or toxicity in the particle and plaque arms were

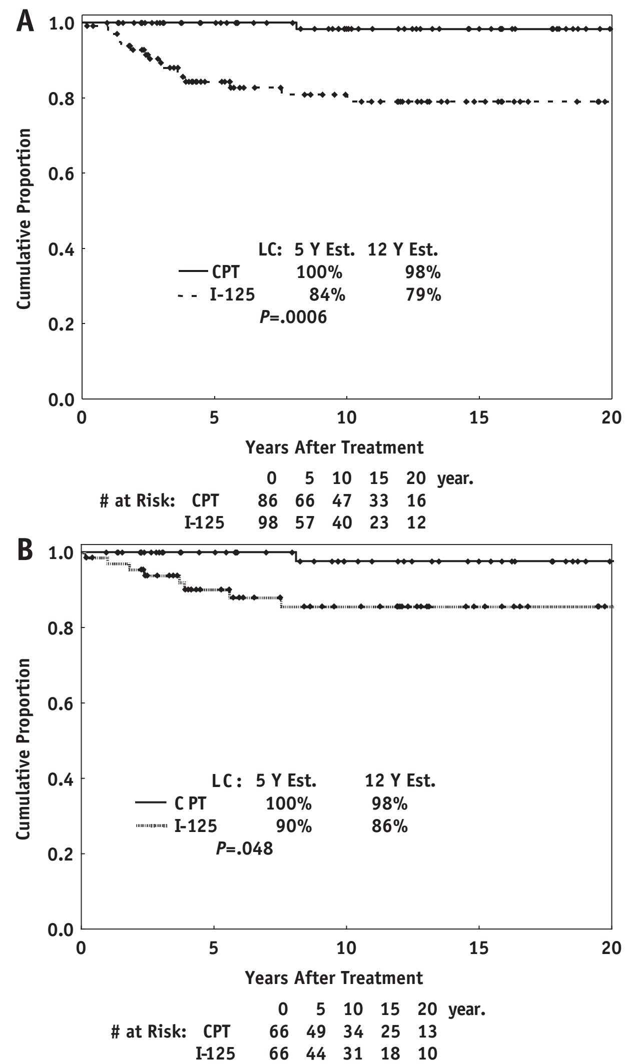

Fig. 1. (a) Kaplan-Meier estimates of local control after CPT with helium-ion and I-125 plaque therapy (log rank: $P = .0006$). (b) Kaplan-Meier estimates of local control after CPT or plaque therapy for tumors ≥2 mm from the optic disc (log rank: $P = .048$). CPT = charged particle therapy.

17% versus 37% and the 5-year rates were 11% versus 22%. For enucleation cases in the particle arm, the median time to enucleation was 53 (range 11-262) months versus 47 (range 2-190) months in the plaque arm, respectively. In the particle therapy cohort, a total of 2 patients were enucleated due to lack of tumor control and 15 due to other complications, mainly neovascular glaucoma. In the plaque arm, a total of 17 patients underwent enucleation due to lack of tumor control and 16 due to other complications. Using multivariate methods, we found CPT was the only significant predictor of prolonged eye preservation (LLR test: $P = .01$) owing to the difference in frequency of enucleation due to local tumor recurrence.

Cause-specific, disease-free, and overall survival

CSS at 12 years for particle versus plaque treatment arm was 80% (95% CI: 69%-87%) versus 76% (CI: 65%-84%)

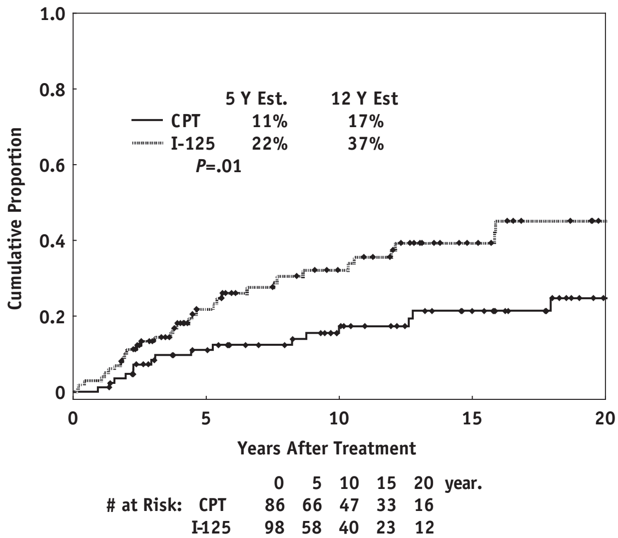

Fig. 2. Kaplan-Meier estimates of enucleation rate after CPT and I-125 plaque therapy (log rank: $P=.01$). CPT = charged particle therapy.

(log rank: $P=.09$), respectively (Fig. 3a). After adjusting for identified independent predictors, maximum tumor diameter ($P<.0001$), and age ($P=.03$), treatment did not have a significant effect on CSS ($P=.14$).

Longer DFS was seen with particle treatment (log rank: $P=.001$) (Fig. 3b). After adjusting for age ($P<.0001$) and tumor diameter ($P=.002$) in a multivariate model (LLR test), particle treatment significantly improved DFS compared with plaque therapy (adjusted: $P=.02$).

Overall survival for the particle versus plaque arm was 86% (CI: 77%-92%) versus 76% (CI: 66%-83%) at 5 years, respectively, and 67% (CI: 55%-76%) versus 54% (CI: 43%-63%) at 12 years, respectively (log rank, $P=.02$). Using Cox multivariate regression model, only increased age ($P<.0001$) and larger tumor diameter ($P=.004$) were independent predictors of decreased OS. After adjusting for these two factors, treatment was not an independent predictor of OS. Table 2 summarizes the multivariate analyses for LC, enucleation, DFS, CSS, and OS.

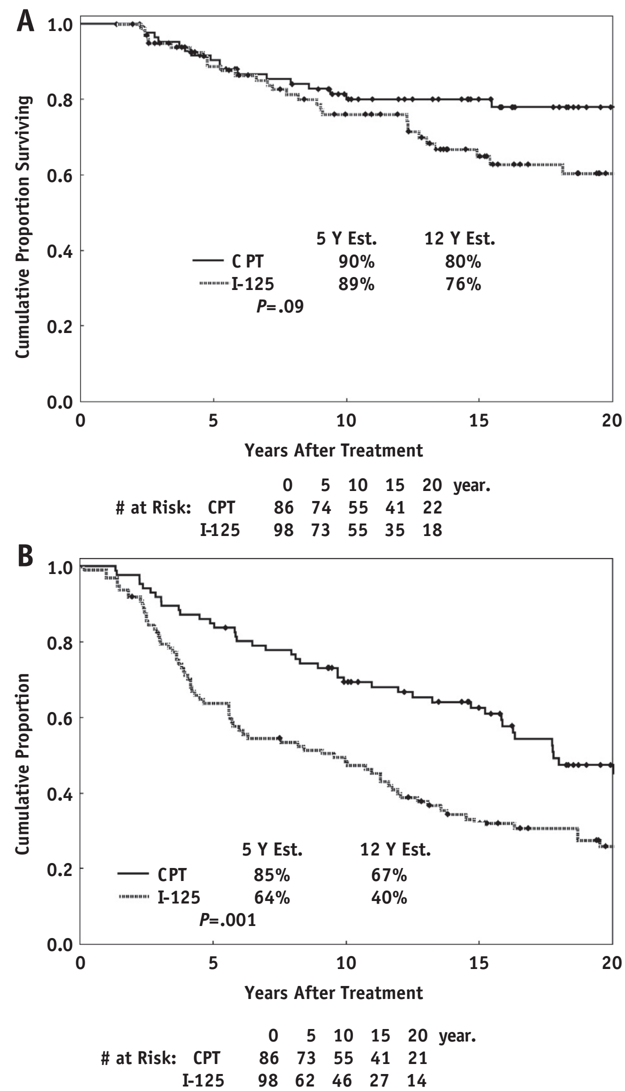

Fig. 3. Kaplan-Meier estimates after CPT and I-125 plaque therapy of (a) cause-specific survival (log rank: $P=.09$) and (b) disease-free survival (log rank: $P=.001$). Cox multivariate regression model shows treatment is a significant predictor of DFS (adjusted: $P=.02$); age and tumor diameter are independent predictors of CSS and DFS. CPT = charged particle therapy; CSS = cause-specific survival; DFS = disease-free survival.

Discussion

Clinical benefits of charged particle therapy

The American Society for Radiation Oncology (ASTRO) recently reviewed published studies of emerging technologies, specifically particle therapy, and stated that there is evidence of clinical benefit particularly for uveal melanomas; however, more prospective clinical trials are needed to analyze the value of CPT (14). Similarly, the US National Institutes of Health/National Cancer Institute published a current national funding effort to encourage CPT research due to the need for studies of the effectiveness of proton and heavier ion therapy (15). Although there is interest in the study of CPT, there are limited particle centers and long-term data available. Our report provides unique evidence of the long-term clinical benefits of CPT, specifically showing that particle therapy resulted in significantly improved LC and eye preservation versus plaque radiation with longer than a 12-year median follow-up in both arms.

The LC rate in the particle versus plaque arm was 100% versus 84% at 5 years, and 98% versus 79% at 12 years, respectively (log rank: $P=.0006$). Importantly, excluding those patients with tumors close to the optic disc (<2 mm), the LC rate in the particle versus plaque arm was 100%

Table 2 Multivariate analysis to identify predictors of outcome

Predictor	Hazard ratio	95% CI	Probability LLR test
LC			
Treatment	8.82	2.03-38.30	.0002
Distance to fovea	0.78	0.64-0.96	.04
Tumor diameter	1.24	1.03-1.48	.02
Enucleation			
Treatment	2.06	1.14-3.70	.01
DFS			
Age	1.05	1.03-1.06	<.0001
Tumor diameter	1.11	1.04-1.18	.002
Treatment	1.54	1.05-2.24	.02
CSS			
Tumor diameter	1.24	1.11-1.38	.0001
Age	1.03	1.00-1.05	.03
OS			
Age	1.07	1.05-1.08	<.0001
Tumor diameter	1.10	1.03-1.18	.004

Abbreviations: CSS = cause-specific survival; DFS = disease-free survival; LLR = likelihood ratio; LC = local control; OS = overall survival.

Final multivariate models using Cox regression model with predictors of LC, enucleation, DFS, CSS, and OS are shown in order of statistical significance based upon the likelihood ratio test.

versus 90% at 5 years and 98% versus 86% at 12 years, respectively (log rank: $P = .048$).

At the time of this study, patients with peripapillary tumors (<2 mm from the disc) were included, and subgroup analysis showed that these patients had poor LC when treated with plaques, as is now well documented (19, 20). Figure 1b displays the results for our patient subset that is more analogous to contemporary patients undergoing plaque treatment (LC 90% at 5 years) and is similar to the plaque therapy outcomes in the multi-institutional Collaborative Ocular Melanoma Study (COMS trial) (LC 89.7% at 5 years) (20).

The relatively poorer outcomes with plaque treatment for peripapillary tumors may be due in part to the mechanical issues with plaque placement, even if notched, at the optic nerve and plaques potentially lifting off the scleral surface. More recent developments in plaque therapy technique have shown improvement in contemporary reported outcomes (19, 20). For example, Badiyan et al (21) have shown plaque tilt >1 mm away from the sclera at time of plaque removal in 27% of cases. Their use of ultrasonography-guided plaque localization and detection of plaque displacement with supplemental transpupillary thermotherapy resulted in improved outcomes, with a 5-year LC rate of 95.4% at 46 months median follow-up.

Local control differences between particle and plaque therapy may be due to radiobiological differences and dose delivery. The RBE of these 2 types of radiation and the higher linear energy transfer of particle beams may be important for LC (22-24). Dose delivery rate variance may result in variable tumor response. Plaque therapy is delivered with a continuous low-dose rate plaque, in our study, over approximately 96 hours. The equivalent total dose of helium particles was delivered in 5 high-dose fractions, with each fraction delivered in <2 minutes. In vitro studies have shown that continuous low-dose rate RT can allow for cellular recovery of radiation injury (sublethal damage repair) (25). The range of biological importance may include 10 cGy per hour to 500 cGy per minute, and hence potentially applicable in melanoma. The radiation survival curve for in-vitro melanoma cell lines suggest that these tumors may be most effectively treated at a high-dose rate (25-27).

Other parameters such as surgical technique and plaque parameters including treatment duration, apex dose rate, and tumor margin did not show an effect on LC in our cohort. The subset sample sizes in this study are limited however, and larger studies are needed to further delineate the relationship between these factors with local control. The surgical technique was consistent and was performed by a single surgeon with extensive experience in both types of surgery for plaque placement and tantalum ring placement. The dose rate itself was consistent with current recommendations (0.60-1.05 Gy/h) (19). In terms of total dose, 70 GyE was selected for both arms as a consistent radiation dose. Studies by Nag et al (19) and Quivey et al (24) reported that, potentially, both the total dose and the dose rate are important for tumor control. It is possible that the total dose of 70 GyE was not adequate for plaques. An early trend of higher local failure in the I-125 arm led the investigators to alter the target volume to include a greater margin and deliver 70 GyE to a point 1 mm above the tumor apex. The tumor apex received approximately 80 GyE, yet these adaptations did not lower the local failure rate in our patient subset. Interestingly, recent work has supported that doses at least as low as 69 Gy to the tumor apex have resulted in similar rates of LC, metastasis-free survival, and OS, and improved visual outcomes, as compared to doses of 85 Gy (28).

The American Brachytherapy Society recommends a tumor I-125 dose of 85 Gy to the apex at a dose rate of 0.60 to 1.05 Gy/h (19). The current recommendation for plaques states that patients with large tumors, or peripapillary/macular tumors, have a poorer visual outcome and lower LC that must be taken into account in the patient decision-making process. Patients with gross extrascleral extension, ring melanoma, and tumor involvement of more than half of the ciliary body are not suitable for plaque therapy. Given the historical nature of this trial, certainly there are differences from current techniques and parameters, importantly the inclusion of peripapillary tumors. Excluding these tumors, the current study outcomes are consistent with overall brachytherapy findings. In a recent review of radiation modalities by Chang and McCannel (12), published brachytherapy studies with an average of 5 years of follow-up ($n = 3868$ patients) showed a weighted average local failure rate of 9.5% (weighted by study sample size). Charged particle therapy studies (n=7043 patients) showed a weighted local failure rate of 4.2%. Overall, particle studies had somewhat larger tumors on average (particle vs plaque weighted mean largest tumor diameter: 13.93 mm vs 11 mm, respectively, and height 5.54 mm vs 4.48 mm, respectively).

The enucleation rate was lower after CPT than after plaque therapy: 11% versus 22% at 5 years and 17% versus 37% at 12 years, respectively, due mainly to differences in tumor control. Enucleation in the particle arm was largely related to neovascular glaucoma (NVG) effects. Of note, our institutional data have shown an overall lower NVG rate for current proton treated patients than our historical helium patient cohort (29). This may be due to changes in patient selection and tumor characteristics, dosing, relative biologic effect of protons versus helium, dose volume changes to anterior and posterior critical structures, use of anti-VEGF therapy, surgical resection and other developments (29-35). The rate of enucleation due to NVG in our current proton patient cohort was 4.9% at 5 years (29). Hence, the overall enucleation rate for particle therapy is expected to be lower in contemporary particle patients than in those treated in this trial. The rate of enucleation is also expected to be lower in contemporary plaque patients than in those in this trial, due to the historical inclusion of peripapillary tumors and improvements in plaque placement techniques. For example, in the COMS trial the enucleation rate after I-125 was 12.5% at 5 years, due mainly to treatment failure (20). Change in visual acuity and other complications are the subject of further study as secondary endpoints for this trial, although the overall difference in enucleation rate between the 2 modalities in this trial was driven by local tumor control differences.

The role of local control in survival for UM patients is controversial. Although some studies have found a relationship, the COMS trial has shown no differences in OS or CSS between enucleation and plaque therapy and showed a 12-year CSS rate of 79% after plaque therapy (36). Consistent with these long-term data, in the current study, 12-year CSS for the particle versus plaque arm was 80% versus 76%, respectively (P=.09). Overall survival rates were not significantly different between treatment arms after adjusting for age. Disease-free survival, accounting for local progression, distant disease or death, was significantly prolonged with particle treatment versus plaque treatment (log rank: P=.001).

Conclusions

Charged particle therapy resulted in higher LC, eye preservation and disease-free survival rates than plaque therapy in choroidal and ciliary body melanoma management, based on the only long-term prospective, randomized clinical trial available to date comparing radiation modalities in UM. The results presented here are consistent with other long-term plaque and particle data. Although currently there are a limited number of particle facilities worldwide, there is a strong developing interest to further invest in particle centers, given the emerging results for CPT in selective disease processes.

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